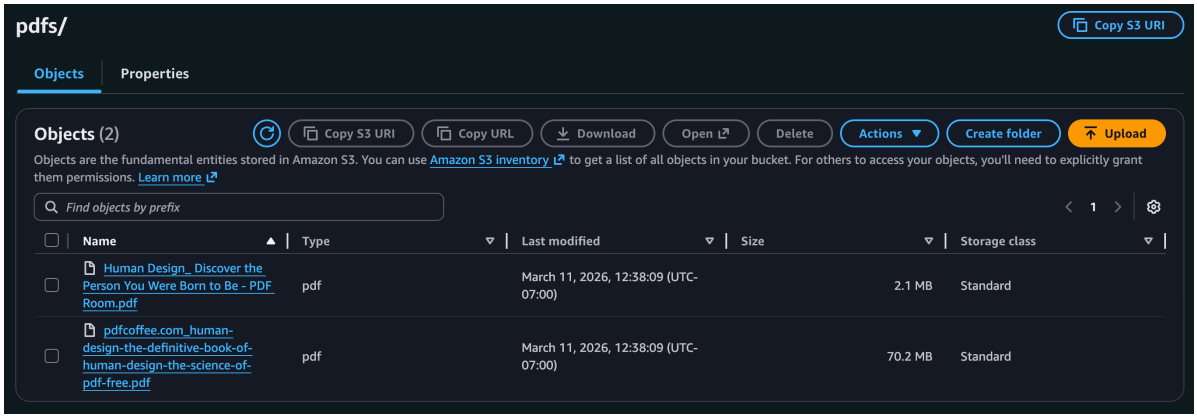
Week1

Phase 1 Status Update

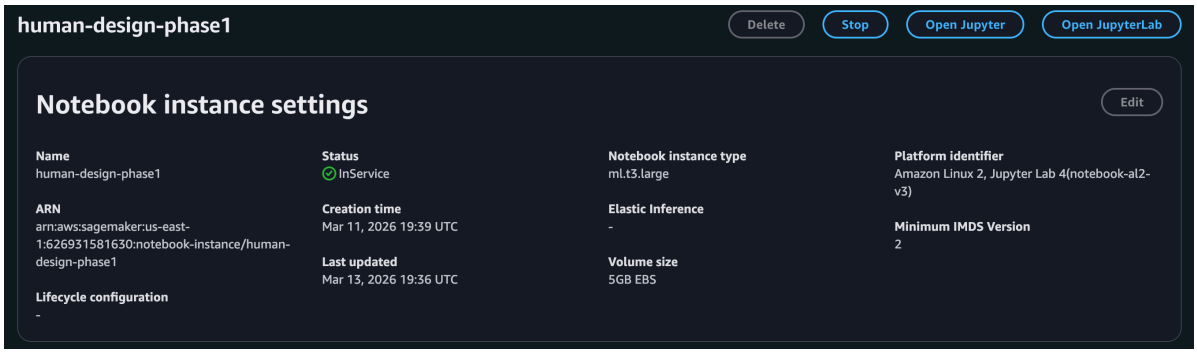
Offline Knowledge Base Construction

- Uploaded 2 Human Design PDF books to S3 bucket (`human-design-knowledge-base`)
  - *The Definitive Book of Human Design* — Ra Uru Hu & Lynda Bunnell
  - *Human Design: Discover the Person You Were Born to Be* — Chetan Parkyn

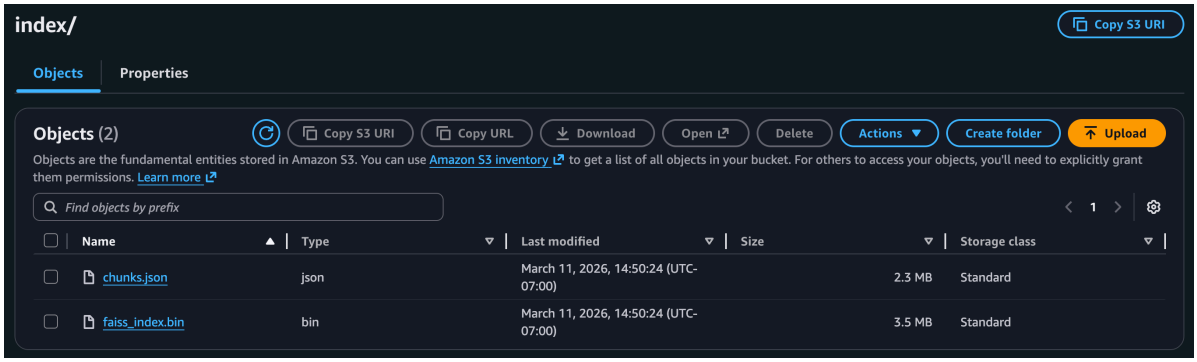


- Built a PDF processing pipeline in SageMaker Notebook:

- Extracted text from PDFs using `PyMuPDF`
- Chunked text into semantically segments
- Generated vector embeddings using `sentence-transformers/all-MiniLM-L6-v2`
- Built a `FAISS` vector index for similarity search



- Stored `faiss_index.bin` and `chunks.json` to S3 for Lambda to consume at runtime



Services used: S3, SageMaker Notebook, FAISS (open-source), sentence-transformers (open-source)

Key design decisions

1. **FAISS over Bedrock Knowledge Bases / OpenSearch Serverless** — The knowledge base is small (~147 chunks). FAISS is free, lightweight, and sufficient for this scale. Since AWS academy don't allow to use **Bedrock**.
2. **Fixed-size chunking with sentence boundary alignment** — Text was split into ~800 character chunks, with boundaries adjusted to avoid breaking mid-

sentence. This ensures each chunk contains complete, readable passages for retrieval.

Next steps

- Phase 2: Vision Extraction Lambda (image → structured chart data)
- Phase 3: RAG Lambda with Docker Container deployment